Software Requirements Specification — PhishGuard AI

Status: Preliminary draft based only on the uploaded project brief. Source-code inspection is pending. Version 0.1, 2026-10-22. IEEE-style organization is used as a guide; no formal compliance is claimed.

1. Introduction

Purpose

Define a testable baseline for PhishGuard AI.

Scope

A browser-based cybersecurity demonstration intended to help users review email and URL phishing-warning analysis. The brief describes HTML, CSS, vanilla JavaScript, local analysis, and browser-local workflows. This is not verified production security software.

Audience

Student team, project guide, evaluator, tester, and maintainer.

Terms

SRS—Software Requirements Specification; UI—User Interface; SIEM—Security Information and Event Management; synthetic data—fabricated demonstration data; heuristic—rule-of-thumb classification.

Organization

Sections cover product context, functional/non-functional requirements, interfaces, data/security, acceptance, and roadmap.

2. Overall description

Product perspective/objectives

Standalone browser-side prototype. Make findings understandable and demonstrate a repeatable local workflow; keep future services distinct from current functionality.

Users

Student operator, evaluator, and cybersecurity-awareness user. Basic browser skills assumed.

Environment

Modern browser with JavaScript. Development tools reported: VS Code and Claude AI; these are not runtime dependencies. Exact OS/browser/version: [record].

Constraints

No backend or live service is assumed. Browser-local storage/export features must be confirmed from source. Client-side code and storage are user-modifiable and not tamper-resistant.

Assumptions/dependencies

Project brief accurately reflects the app; inputs are synthetic or authorized; browser APIs used by the code are available. Unknown implementation details remain placeholders.

3. Functional requirements

Priority is provisional. Requirement statuses below mean described in the brief, not verified in source.

ID	Requirement	Priority	Input	Behavior	Output	Acceptance criteria	Status
PhishGuard-AI-FR-01	The application shall support email input/analysis.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified

ID	Requirement	Priority	Input	Behavior	Output	Acceptance criteria	Status
PhishGuard-AI-FR-02	The application shall support url extraction.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-03	The application shall support url structure analysis.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-04	The application shall support suspicious-pattern detection.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-05	The application shall support explainable risk scoring.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-06	The application shall support evidence display.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-07	The application shall support awareness training.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-08	The application shall support analysis history.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-09	The application shall support report export.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-10	The application shall support input validation.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified
PhishGuard-AI-FR-11	The application shall support privacy controls.	Medium (provisional)	user-provided email text/URLs and synthetic scenarios	Perform local processing and label simulation clearly.	UI finding/output	Execute linked test and record evidence.	Described in brief; not source-verified

4. Non-functional requirements

ID	Requirement	Verification	Status
NFR-01	Usability: plain-language labels and explanations	Walkthrough	Proposed
NFR-02	Performance: no noticeable freeze in core demo	Measure on declared device; no numeric claim	Not tested
NFR-03	Reliability: malformed input does not crash UI	Negative tests	Proposed
NFR-04	Maintainability: separate UI, rules, scoring, persistence where practical	Source review	Proposed
NFR-05	Portability: core demo runs in modern browser without install	Browser matrix	Described; verify
NFR-06	Accessibility: keyboard, focus, labels, contrast	Manual review	Unverified
NFR-07	Privacy: explain local data and inspect network behavior	Storage/network inspection	Unverified
NFR-08	Security: safely render untrusted input; bound inputs	Code review and tests	Proposed
NFR-09	Error handling: actionable errors, no false-success messages	Negative tests	Proposed

ID	Requirement	Verification	Status
NFR-10	Data integrity: reset/export behavior documented	Persistence/export tests	Unverified
NFR-11	Offline operation: core logic works without network if no remote dependency	Disconnect and test	Described; verify

5. External interfaces

UI: actual screens/control names [map from source]. Browser APIs: localStorage/Blob/download only if source confirms. Local storage is not encrypted. Export format, filename, and data fields: [verify]. Future integrations: external AI, reputation/threat intelligence, SPF/DKIM/DMARC checks, malware scanning, and live-site analysis; each needs separate interface/security requirements.

6. Data requirements

Document actual record schemas, validation, retention, deletion, synthetic labels, and report fields from source. Minimize stored user content. Use synthetic fixtures and label them visibly. No schema is confirmed here.

7. Security requirements

Threats include malformed input, stored XSS, misleading scores, local data exposure, report disclosure, and confusion between demo and live service. Trust boundaries: input→parser/rules→UI; app↔browser storage; browser→download. Treat input as untrusted, use safe text rendering, avoid eval/unsafe HTML insertion, and validate size/structure. These are review requirements, not verified controls. Keep phishing results explicitly heuristic and non-guaranteed. Do not imply browser storage provides confidentiality or server-side enforcement.

8. Acceptance and traceability

Acceptance requires execution of linked test cases, recording actual results/evidence, and resolving or formally accepting defects. Current acceptance: **Not evaluated**.

9. Constraints and future enhancements

Current constraints: browser-only, local rules, synthetic/authorized inputs, no verified live external data or production controls. Future: external AI, reputation/threat intelligence, SPF/DKIM/DMARC checks, malware scanning, and live-site analysis. Plan integrations separately with authorization, privacy review, and tests.